

Section 4 - Visualization; Simulation & Particle Distance



This video covers simulation and particle distance. I am using the Lesson linked before the video.

Lecture 11 | Simulation

- TABLECLOTH + SIMULATION
- Closed Library→ Arrow at top right OR Hotkey[Y]
- 3D window only→ button in the bottom right. For this file you only need the 3D window.
- Simulation on→ click the down arrow button in the 3D window here.
 - Hotkey for simulation is the Spacebar.
 - Change shortcut in Settings if necessary
- Simulation→ tablecloth falls on the table
 - 1-Down arrow in the toolbar turns blue
 - 2- the fan on your computer may start running.
 - Simulation is on,computer is making constant calculations to apply gravity to the properties of your fabric.
 - Requires a lot of computing power.
 - TIP; turn simulation on to check drape, turn off when fabric stops moving
- Simulation→ main selection tool in 3D[Select/Move Tool] changes to a hand.
 - Select pattern piece to move/drape in 3D window.
- DEPTH IN CLO
- Pull things up and down or left and right.
- Cannot push or pull toward or away from you.
 - pull toward yourself; rotate view so that "toward" becomes left or right.
- 3D world everything is mesh.
 - garment as well as any avatars.
 - Display views to mesh mode.
 - CLO mesh is triangular by default; allows for softer curves when fabrics drape over the human form.
 - Can be changed in your User Settings or manually by pattern piece in the property editor.
 - alternative is a quad or rectangular mesh, which is the standard for hard 3D modeling software.
 - Mesh objects can pass THROUGH each other.
 - 3D garment mesh & Avatar mesh entangled; called COLLISION
 - TIP; Pulling up and away from the avatar or other garments.

- EG. even out my tablecloth, IRL, I might pull this end down. CLO; gently pull it up and over, so as not to pull it directly into my table.
 - Practice to note how hard or easy to pull
- SIMULATION GAME
 - Training game for practicing moving with simulation on.
 - Remember; you cannot pull things toward or away from you with simulation on.
 - Check your perspective by rotating around 3D window.
 - Depth perception is a bit off in 3D; not as many cues as there are in the real world→ takes some getting used to.
- Simulation off will “pause” the game; freezing your patterns
- Click on them in this state→ gizmo.
 - Click and drag them; equivalent of using the yellow square on your gizmo.
 - Learn about gizmo in Avatar and Arrangement lesson.
- The purpose here is to move things using hand tool while simulation on.
 - If paused to rotate and would like to continue, click on your pattern and hold before turning simulation back on with Spacebar.
 - Allows you to hold the pattern when simulation comes back on. Otherwise, your pattern will fall to the ground where it is.
- SIMULATION MODES
 - GPU+SIMULATION
 - 3 kinds of simulation
 - Click and hold simulation icon in toolbar→ drop-down menu for all modes
 - NORMAL SIMULATION; default working simulation. Click to Start; arrow turns solid blue
 - Speed should be relatively fast with garments in low resolution.
 - Speed depends on the power of your computer; working in real time and my garment moves similarly to the real world.
 - The more things you have in the 3D window, the more calculation CLO must do and the slower the simulation runs.
 - FAST (GPU). when Active; BLUE arrow outline as the icon.
 - Option may not show up if;
 - you do not have a dedicated graphics card
 - using CLO on Mac Computers.
 - GPU simulation means simulation speed is running off your GPU rather than CPU
 - Fastest simulation
 - Useful especially with large garments [bridal or home goods]
 - Used during draft phase of garment; affects the collision thickness of pattern pieces
 - Not ideal to use on garments with many layers.
 - FITTING (ACCURATE FABRIC) simulation.
 - Final garments in high resolution.
 - Most accurate simulation for fitting and drape.
 - When active; icon is double red arrows with red line underscore

- Default mode when for Hi-Res Garment button.
- Ensures the drape of the fabric represents the accurate retention of the garment;
 - It falls, but also springs back as it would IRL.
- Do not pull/adjust; allow everything time to drape/relax on its own
- when simulating, wait for garment to stop moving, then turn simulation off.

Lecture 12 | Particle Distance

- Earlier discussion; everything in 3D is made of mesh.
- COLUMNS FILE
- Square patterns draped over columns. Note corners showing through.
 - Has to do with the particle distance.
 - Corners of column at holes in the triangles of the mesh→ column show through.
- "Resolution" in CLO, most important aspect is Particle Distance.
 - Average distance between the vertices of the mesh.
 - Default working particle distance in CLO is 20; each vertex is 20 millimeters apart.
 - A property of the pattern pieces.
 - Change on individual pattern pieces in property editor.
 - Particle Distance just under Simulation Properties.
- Default for High Resolution is 5mm.
 - Simulate; fill in the holes, wrinkles will become more realistic and defined.
- PARTICLE DISTANCE FILE
- Garment with different patterns in different particle distance.
 - Skirt is set to 75' looks like crumpled paper.
 - Bodice is 20, neck band is 5.
 - Simulation on skirt incredibly fast.
 - Particle distance affects the simulation speed.
 - High PD, fewer vertices, fewer calculations for gravity.
 - Why default particle distance is 20.
 - Get a decent looking garment, keeping the simulation speed fast while working.
 - No GPU simulation option? Very large pattern pieces?
 - PD above 20.
 - Set skirt at 20 now & check simulation speed.
 - Garment still drapes plenty fast, though neck trim is set to 5.
 - Normal; set pattern pieces that are smaller than your hand in width or length to lower particle distance, even while working.
 - Shouldn't slow your simulation down
 - Help smaller pieces not collide with other patterns.
 - Less mesh & mesh is too open; small pattern pieces more likely to entangle with other patterns.
 - 20 & simulate = collision

- PD Less than 5→ pop-up warning.
 - Ignore & continue
 - Only for very small/thin
 - EG. spaghetti strap, etc
- Set all my pattern pieces to 5 Particle Distance.
 - Garment looking kind of pebbly.
 - Always need to simulate after changing to re-drape.
 - Lift skirt; simulation speed much slower.
 - Also; elastic smocking waistband is significantly different
 - Anything with gathers, wrinkles, elastic; look much better once you change the particle distance.
 - Check then change the particle distance back and keep working.
- PD does not change drape of the fabric.
 - Drape of the fabric is determined by fabric's physical properties.
 - Unrealistic edge if the mesh is too big.
 - Do all virtual fittings when the garment is in high resolution and use our most accurate simulation mode ; FITTING (ACCURATE FABRIC).

HI-RES/LOW-RES GARMENT

- Factors that create a Hi-Res/Low-Res Garment
 - Particle Distance
 - Collision
 - "Additional Thickness-Collision" or Collision Thickness.; Default is 2.5mm.
 - Think "buffer" or "invisible forcefield"
 - Avatar Skin Offset
 - Default is 3mm
 - "Additional Thickness-Render" does not relate. Default is 0
 - All can be changed in Property Editor
 - Select Pattern[s]; Simulation Properties→ change PD & Collision
 - Select Avatar; Surface→ change Skin Offset
- Garment into Hi-Res; Hi-Res Garment Tool.
 - 3D toolbar, the half t-shirt icon with an arrow pointing up.
 - Pop-up. Default Settings
 - All patterns >5 particle distance down to 5.
 - Not affect any pattern pieces that are already smaller than 5.
 - Collision thickness
 - The default is down from 2.5 to 1mm.
 - Between patterns & avatar, and patterns & each other.
 - Skin offset will be brought down to 0.
 - Ok to set at 1; won't be noticeable.
 - Settings save with the program.
 - Experiment & use your own settings
 - Simulation; default for Hi-Res is FITTING (ACCURATE FABRIC)

- This will be slower than Normal and if your computer is very basic,
 - Ok to leave at Normal
 - Must use for active, swim, intimate wear
 - Close fit to body
 - OK→ few seconds to load all those settings.
 - Simulate to drape in hi-res.
- Want to Edit Hi-Res Garment→ Low Res Garment Tool
 - Bring settings back to working settings.
 - Be mindful of Pattern Pieces you want to stay Hi-Res
 - This will avoid things like the neck trim on the dress getting entangled.
 - Select Pattern Pieces & Change PD in Property Editor